

① Define upper and lower Triangular matrix.

→ A square matrix in which all the elements above the main diagonal are zero then it is called a lower triangular matrix.

Eg
$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{bmatrix}$$

→ Similarly, A square matrix with all elements below diagonal as zero is called upper triangular matrix.

② Find x if A is a singular matrix

$$A = \begin{bmatrix} 3-x & 2 & 2 \\ 2 & 4-x & 1 \\ -2 & -4 & -1-x \end{bmatrix}$$

Soln

Singular matrix $|A| = 0$

$$(3-x)[(4-x)(-1-x) - (-4)] - 2[(2)(-1-x) - (-2)]$$

$$+ 2[(-8) - (-8 + 2x)] = 0$$

$$(3-x)[(-4 - 4x + x + x^2) + 4] - 2[(-2 - 2x) + 2]$$

$$+ 2[-8 + 8 - 2x] = 0$$

$$(3-x)[x^2 - 3x] + 4x - 4x = 0$$

$$(3-x)[x(x-3)] \quad \boxed{x = 0, 3, -3}$$

3. $A^2 - 3A + 9I$ where $A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ -3 & 1 & 2 \end{bmatrix}$ 25BCON1077

$$A^2 = A \times A$$

$$A^2 = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ -3 & 1 & 2 \end{bmatrix} \times \begin{bmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ -3 & 2 & 2 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} -12 & -5 & 11 \\ 11 & 4 & 1 \\ -7 & 14 & -6 \end{bmatrix}$$

$$3A = \begin{bmatrix} 3 & -6 & 9 \\ 6 & 9 & -3 \\ -9 & 3 & 6 \end{bmatrix}$$

$$A^2 - 3A + 9I = \begin{bmatrix} -12 & -5 & 11 \\ 11 & 4 & 1 \\ -7 & 14 & -6 \end{bmatrix} - \begin{bmatrix} 3 & -6 & 9 \\ 6 & 9 & -3 \\ 9 & 3 & 6 \end{bmatrix}$$

$$+ \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} -6 & 1 & 2 \\ 5 & 4 & 4 \\ 2 & 8 & -3 \end{bmatrix}$$

Define Symmetric and Skew-Symmetric Matrix. 25BCL01877

- A matrix is called symmetric if it's equal to transpose of itself and is also a square matrix.
- A square matrix is called as skew-symmetric matrix if the transpose of matrix is negative of matrix.

$$\begin{array}{l} \text{Symm} \rightarrow A = A^T \\ \text{Skew-Symm} \rightarrow A = -A^T \end{array}$$

5 Define Rank-nullify theorem.

This theorem relates the number of linearly independent rows (rank) and number of solutions (nullify) for any matrix A.

$$\text{rank}(A) + \text{nullify}(A) = \text{No. of columns of } A$$

[Section-B]

1 Find the inverse of following matrices by row transformation.

$$A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

Soln

$$A = IA$$

$$\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & -14 & 16 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_2$$

$$\begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\boxed{\text{Rank} = 2}, n = 3$$

$$\begin{aligned} x + 3y - 2z &= 0 \\ -7y + 8z &= 0 \end{aligned}$$

$$y = \frac{8}{7}z$$

$$x + 3\left(\frac{8}{7}z\right) - 2z = 0$$

$$x = -\frac{10}{7}z$$

$$x = -\frac{10}{7}k, y = \frac{6k}{7}, z = k$$

$$\begin{bmatrix} 1 & 3 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} A$$

$$R_1 \rightarrow R_1 - 3R_2 - 3R_3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \text{ Ans}$$

2. Solve the following homogeneous system of equations.

$$x + 3y - 2z = 0$$

$$2x - y + 4z = 0$$

$$x - 11y + 14z = 0$$

$$\begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 4 \\ 1 & -11 & 14 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

Applying row operations

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 4 \\ 1 & -11 & 14 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 0 \\ 0 & -14 & 16 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_1$$

$$\begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\rho(A) = 2 < 3$$

$$\boxed{\text{rank} = 2}, x = 3$$

$$\begin{aligned} x + 3y - 2z &= 0 \\ -7y + 0z &= 0 \end{aligned}$$

$$y = 0/7z$$

$$x + 3\left(\frac{0}{7}z\right) - 2z = 0$$

$$x = -10/7z$$

$$x = -\frac{10x}{7}, y = \frac{6x}{7}, z = x$$

3. Find the rank of matrix A = $\begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 3 \\ 1 & 3 & 4 & 1 \end{bmatrix}$

solving by row operation method

$$R_2 \rightarrow R_2 - 3R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$\Rightarrow$

$$\begin{bmatrix} 1 & 3 & 4 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

$$\boxed{\text{rank} = 2}$$

(~~non~~ non-zero rows)

Section-C

25/01/2022

1. Test for consistency of the following system of equations and if possible, find the soln.

$$x + y + z = 8$$

$$2x + 3y + 2z = 19$$

$$4x + 2y + 3z = 23$$

Soln

$$C = [A : B]$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 8 \\ 2 & 3 & 2 & 19 \\ 4 & 2 & 3 & 23 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - 4R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 8 \\ 0 & 1 & 0 & 3 \\ 0 & -2 & -1 & -9 \end{array} \right]$$

$$R_3 \rightarrow R_3 + 2R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 8 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & -1 & -3 \end{array} \right]$$

$$x + y + z = 8$$

$$y = 3$$

$$z = 3$$

$$x = 2$$

no contradictory row like $[000]C$ with $C \neq 0$ so the system is consistent and has a unique soln.

2. Test for consistency of the following system of equations and if possible, find their soln. Q5B CON 1072

$$x + y + z = 6, \quad x + 2y + 3z = 14, \quad x + 4y + 7z = 30.$$

$$C = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 14 \\ 1 & 4 & 7 & 30 \end{array} \right]$$

$$R_2 \rightarrow R_2 - R_1, \quad R_3 \rightarrow R_3 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 3 & 0 & 24 \end{array} \right]$$

$$R_3 \rightarrow R_3 - 3R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$y + 2z = 8 \Rightarrow y = 8 - 2z$$

$$x + (8 - 2z) + z = 6$$

$$x = z - 2$$

$$\boxed{x = k - 2, \quad y = 8 - 2k, \quad z = k}$$

Test for consistency of the following system of equations and if possible, find the soln. R5BLON1077

$$2x - 3y + 7z = 5, \quad 3x + y - 3z = 13, \quad 2x + 15y - 47z = 32$$

$$C = [A:B]$$

$$C = \begin{bmatrix} 2 & -3 & 7 & : & 5 \\ 3 & 1 & -3 & : & 13 \\ 2 & 15 & -47 & : & 32 \end{bmatrix}$$

$$\text{Swap } R_1 \leftrightarrow R_2$$

$$\begin{bmatrix} 3 & 1 & -3 & : & 13 \\ 2 & -3 & 7 & : & 5 \\ 2 & 15 & -47 & : & 32 \end{bmatrix}$$

$$R_2 \rightarrow 3R_2 - 2R_1 \text{ and } R_3 \rightarrow 3R_3 - 2R_1$$

$$\begin{bmatrix} 3 & 1 & -3 & : & 13 \\ 0 & -11 & 27 & : & -11 \\ 0 & 55 & -135 & : & 70 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 5R_2$$

$$\begin{bmatrix} 3 & 1 & -3 & : & 13 \\ 0 & -11 & 27 & : & -11 \\ 0 & 0 & 0 & : & 15 \end{bmatrix}$$

$$0x + 0y + 0z = 15 \\ 0 \neq 15$$

This is contradiction
Inconsistent and
have no solution.

~~9/15/26~~